1.

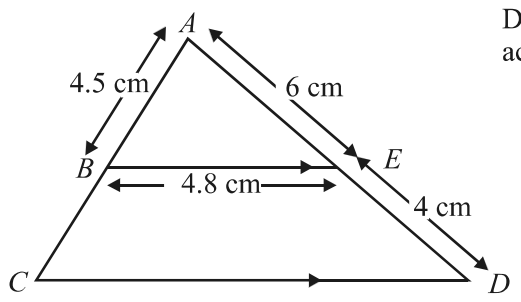


Diagram **NOT** accurately drawn

BE is parallel to *CD*.
AE = 6 cm, *ED* = 4 cm, *AB* = 4.5 cm, *BE* = 4.8 cm.

(a) Calculate the length of *CD*.

.....cm
(2)

(b) Calculate the perimeter of the trapezium *EBCD*.

.....cm
(2)
(Total 4 marks)

2.

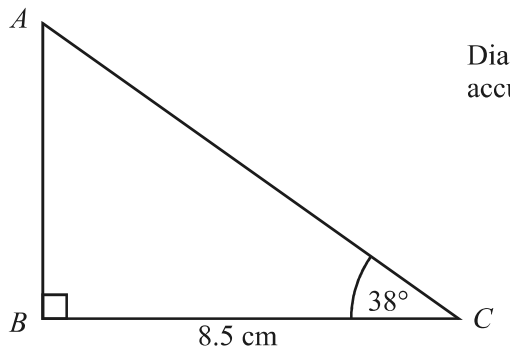


Diagram **NOT** accurately drawn

The diagram shows triangle *ABC*.
BC = 8.5 cm.
Angle *ABC* = 90°.
Angle *ACB* = 38°.

Work out the length of *AB*.
Give your answer correct to 3 significant figures.

..... cm
(Total 3 marks)

3.

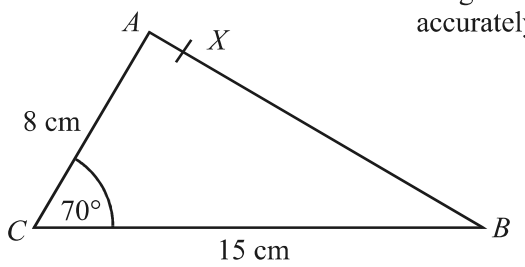


Diagram **NOT** accurately drawn

In triangle *ABC*,
AC = 8 cm,
CB = 15 cm,
Angle *ACB* = 70°.

(a) Calculate the area of triangle *ABC*.
Give your answer correct to 3 significant figures.

..... cm^2

(2)

X is the point on *AB* such that angle *CXB* = 90°.

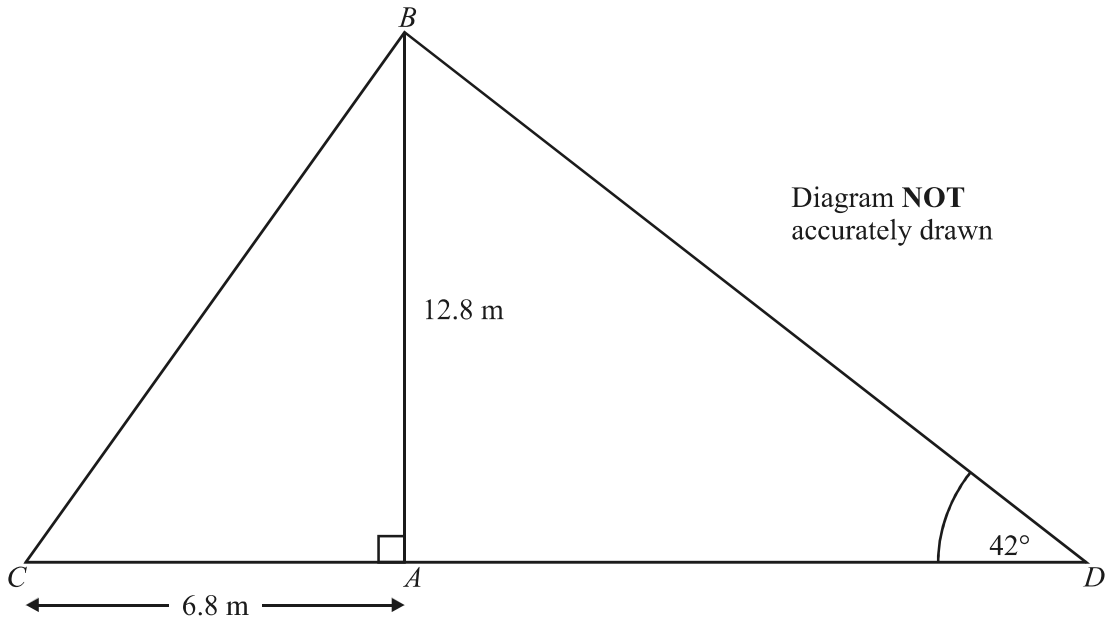
(b) Calculate the length of *CX*.
Give your answer correct to 3 significant figures.

..... cm

(4)

(Total 6 marks)

4. The diagram represents a vertical flagpole, *AB*.
The flagpole is supported by two ropes, *BC* and *BD*, fixed to the horizontal ground at *C* and at *D*.



AB = 12.8 m.
AC = 6.8 m.
Angle *BDA* = 42°.

(a) Calculate the size of angle *BCA*.
Give your answer correct to 3 significant figures.

……°

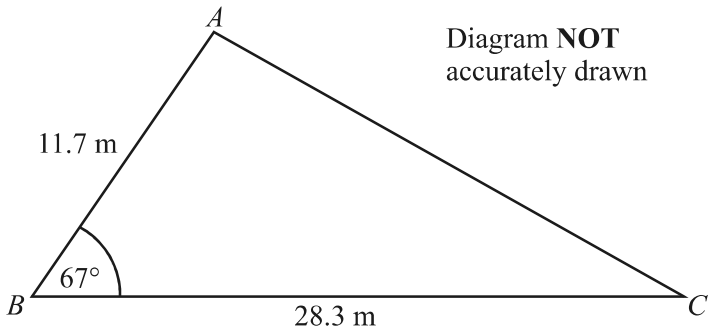
(3)

(b) Calculate the length of the rope *BD*.
Give your answer correct to 3 significant figures.

…… m

(3)
(Total 6 marks)

5.



AB = 11.7 m.
BC = 28.3 m.
Angle *ABC* = 67°.

(a) Calculate the area of the triangle *ABC*.
Give your answer correct to 3 significant figures.

…… m^2

(2)

(b) Calculate the length of AC.
Give your answer correct to 3 significant figures.

…… m

(3)

(Total 5 marks)

6. The depth, D metres, of the water at the end of a jetty in the afternoon can be modelled by this formula

$$D = 5.5 + A \sin 30(t - k)°$$

where

t hours is the number of hours after midday,
A and k are constants.

Yesterday the low tide was at 3 p.m.
The depth of water at low tide was 3.5 m.

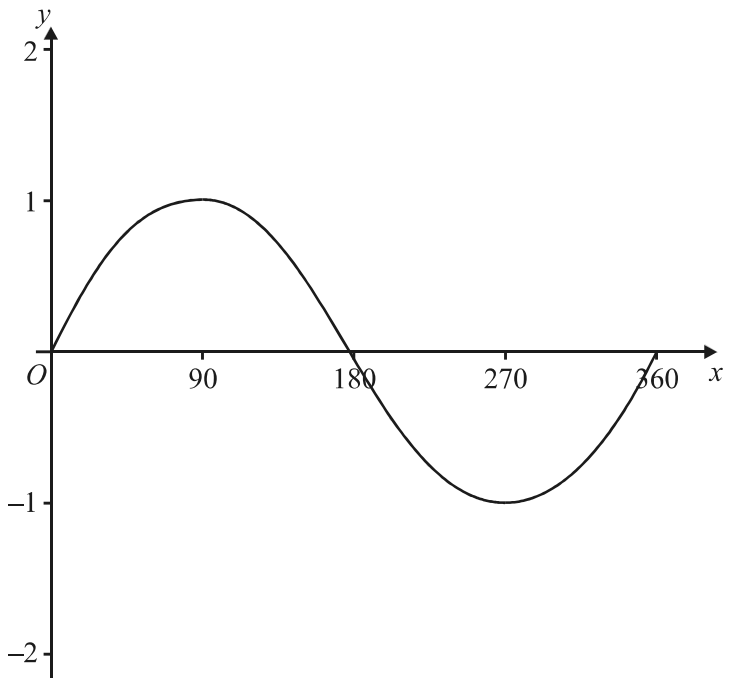
Find the value of A and k.

A = ...

k = ...

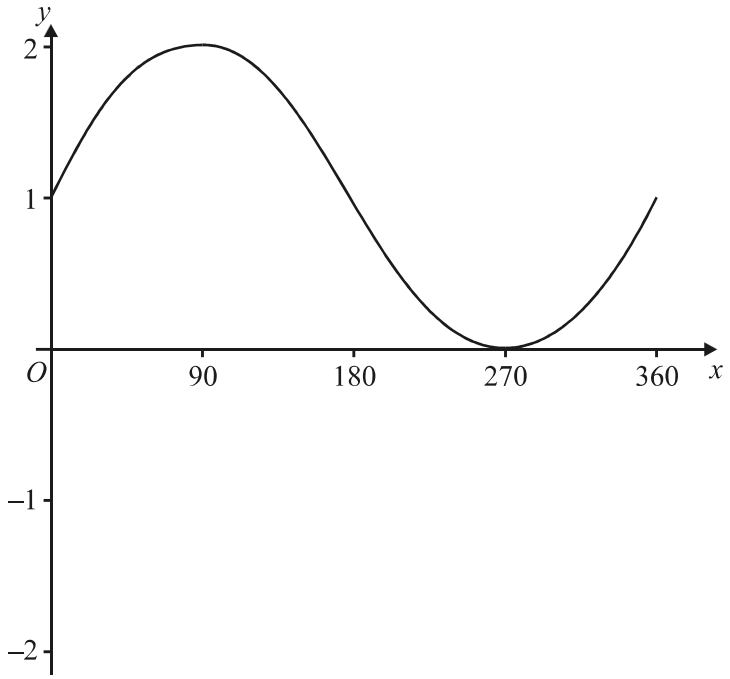
(Total 4 marks)

7. A sketch of the curve $y = \sin x°$ for $0 \le x \le 360$ is shown below.



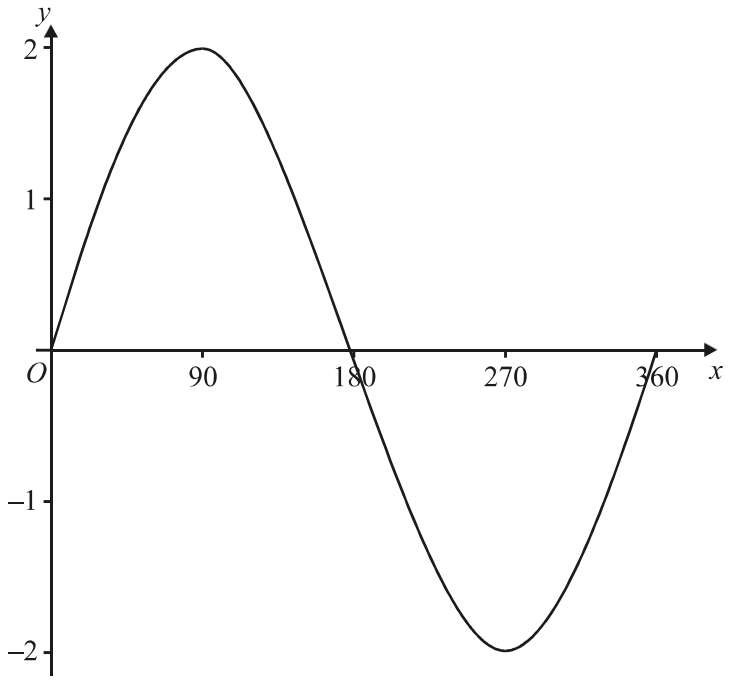
(a) Using the sketch above, or otherwise, find the equation of each of the following two curves.

(i)



(i) Equation y = ...

(ii)



(ii) Equation y = ...

(2)

(b) Describe fully the sequence of two transformations that maps the graph of $y = \sin x°$ onto the graph of $y = 3 \sin 2x°$

.....

.....

.....

.....

(3)
(Total 5 marks)

8.

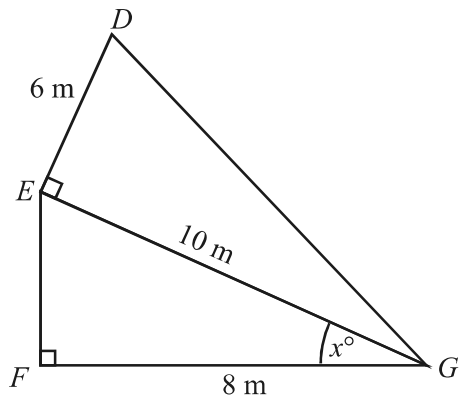


Diagram **NOT** accurately drawn

DE = 6m.
EG = 10 m.
FG = 8 m.
Angle DEG = 90°. Angle EFG = 90°.

(a) Calculate the length of DG.
Give your answer correct to 3 significant figures.

..... m
(3)

(b) Calculate the size of the angle marked x°.
Give your answer correct to one decimal place.

.....°
(3)
(Total 6 marks)